

(1)

HW # 9 Idea

① Computing Jacobians numerically

a) Need a function we can do by hand $f: \mathbb{R}^5 \rightarrow \mathbb{R}$

b) Introduce idea of Symmetric Differences

$$\left. \frac{\partial f}{\partial x_i} \right|_{x^*} \approx \frac{f(x^* + \delta e^i) - f(x^* - \delta e^i)}{2\delta}$$

where $e^i =$ natural basis vector

c) Introduce regression approach

$$f(x^* + \Delta x) \approx f(x^*) + \left. \frac{\partial f}{\partial x} \right|_{x^*} \cdot \Delta x$$

$$= b + C \Delta x$$

$$\neq b + C \Delta x$$

(2)

Choose $\{(\Delta x_k) \mid 1 \leq k \leq N\}$ "randomly"

and small $\Delta x = (\delta x_1, \delta x_2, \dots, \delta x_n)$

Measure $y_k = f(x^* + \Delta x_k)$

Fit model $y = b + c_1 \delta x_1 + c_2 \delta x_2 + \dots + c_n \delta x_n$

$$= \begin{bmatrix} 1 & \delta x_1 & \delta x_2 & \dots & \delta x_n \end{bmatrix} \begin{bmatrix} b \\ c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$$Y_k = \begin{bmatrix} y_1 \\ \vdots \\ y_k \end{bmatrix}$$

$$Y_k = \underbrace{\begin{bmatrix} 1 & (\delta x_1)_1 & \dots & (\delta x_n)_1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & (\delta x_1)_k & \dots & (\delta x_n)_k \end{bmatrix}}_{\text{full rank}} \begin{bmatrix} b \\ c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$$\Rightarrow \text{Solve for } c_1, \dots, c_n \text{ (2b)} \Rightarrow \left. \frac{\partial f}{\partial x} \right|_{x^*} \approx C$$

(3)

(d)

Give them a function that is challenging to do by hand, and we do it numerically, using both methods.

(e) Add a bit of noise to the function measurement. Then compare

Symmetric Differences vs Regression